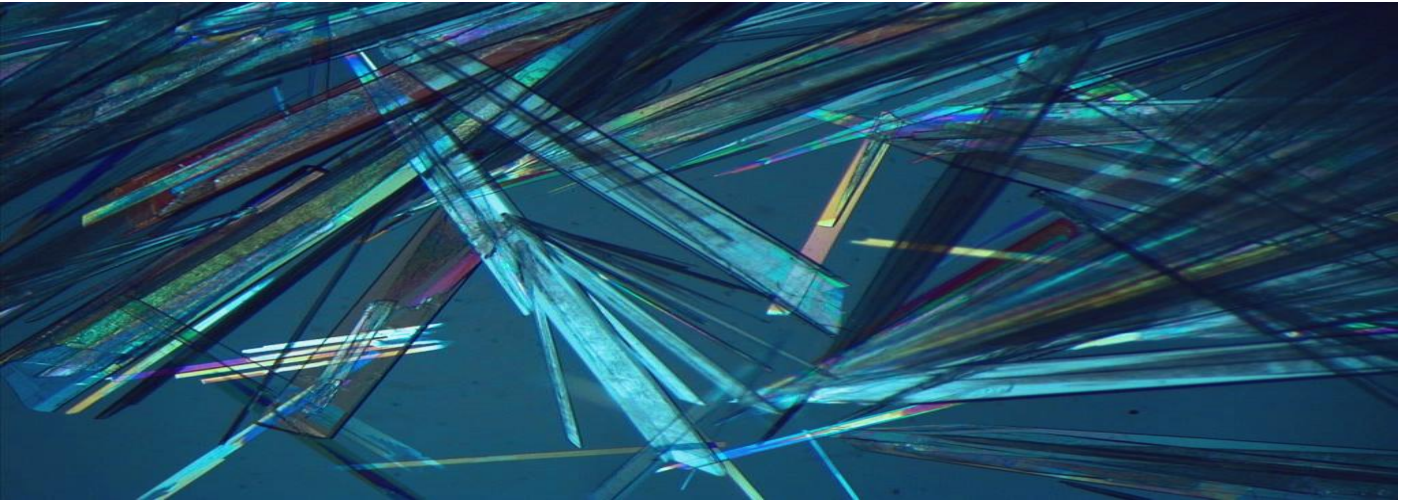



Advanced Materials (AdMat)
in Crystal Engineering

Scope

Heinrich Heine
Universität Düsseldorf

- What is Crystal Engineering and why do we study it?
- Background and basics
- Intermolecular Interactions
- Crystal design principles
- Polymorphs
- Multi-component crystals
- Structure-property relation
- Functional materials and their properties
- Concepts and strategies to develop new materials
- Boundaries

Goals:

- ✓ Understanding of CE principles
- ✓ Critical assessment of possibilities & perspectives
- ✓ Structural design to function transfer
- ✓ Realistic overview of the potential for developing functional materials

Literature & Software

- G. R. Desiraju, J. J. Vittal, A. Ramanan, Crystal Engineering: A Textbook, 2011, World Scientific Publishing,
- E. R. T. Tiekink, J. J. Vittal, M. J. Zaorotko (Eds.), Organic Crystal Engineering. Frontiers in crystal engineering, 2010, Wiley Publishing
- J. Bernstein, Polymorphism in Molecular Crystals, 2008 International Union of Crystallography, Oxford Press
- E. R. T. Tiekink, J. Zukerman-Schpector (Eds.), Multi-component crystals. Synthesis, concepts, function, 2018 De Gruyter
- Scientific publications (online search: Google Scholar, SciFinder Database, Scopus and Web of Science)

- **ILIAS**: AdMat-CE group

- CSD-Database (Cambridge Structural Database): Web CSD with HHU license

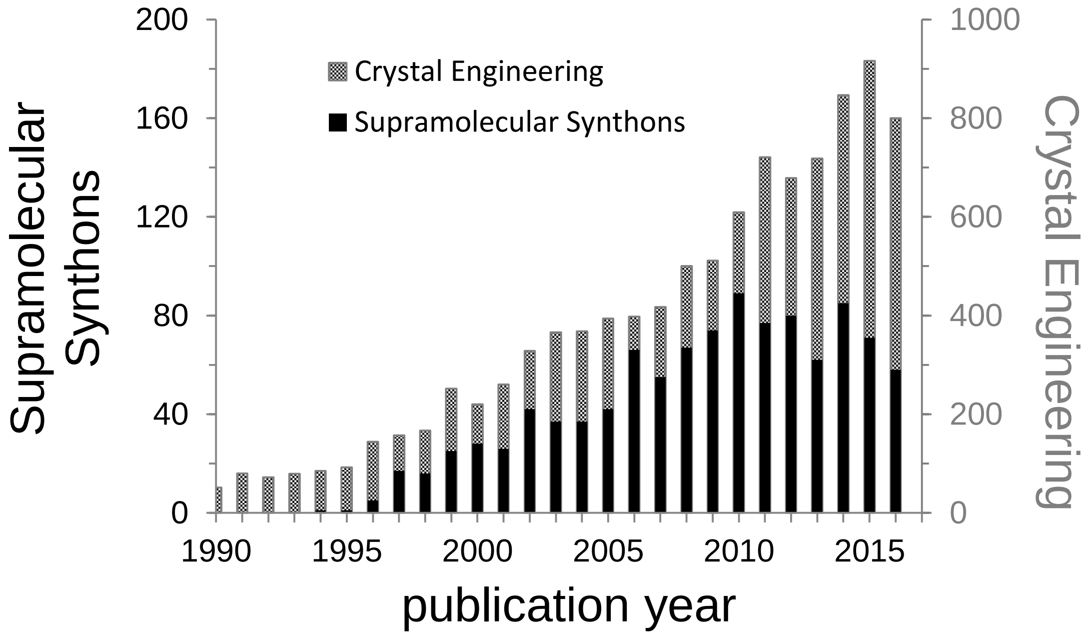
  >> registration with HHU email > download Mercury or full package (10 GB)
  Site/Customer Number: 531
  Licence Key: D9A73E-B5F993-4C1196-B4B3BE-2E256C-F4084F

- Mercury Software from CCDC (Cambridge Crystallographic Data Centre) https://www.ccdc.cam.ac.uk/

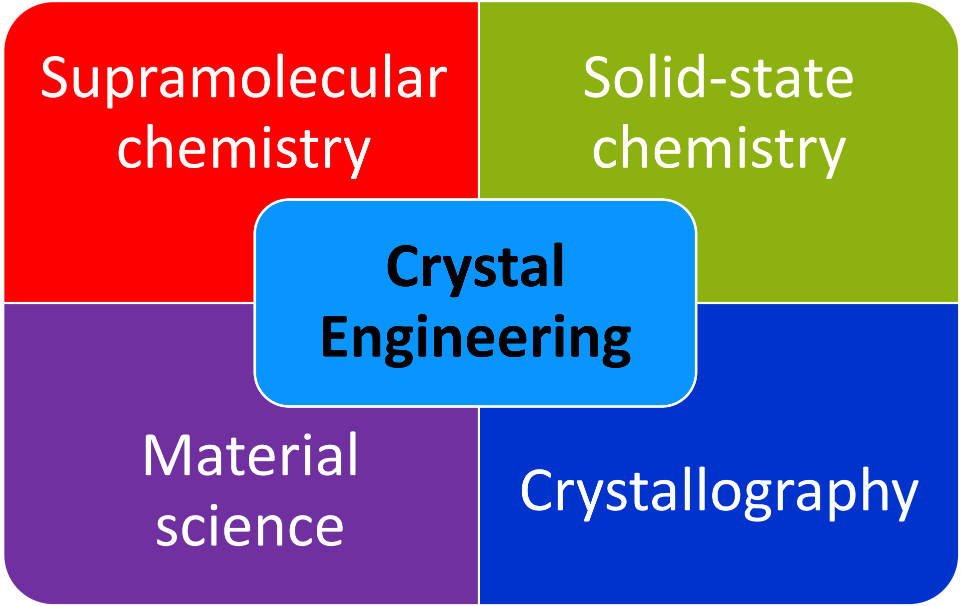
Crystal Engineering: young field with many perspectives

hhu
Heinrich Heine
Universität Düsseldorf

A survey of articles on the topic of CE   from 1990 to 2016



Is CE Inorganic Chemistry?

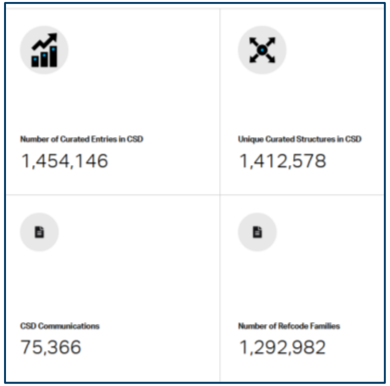


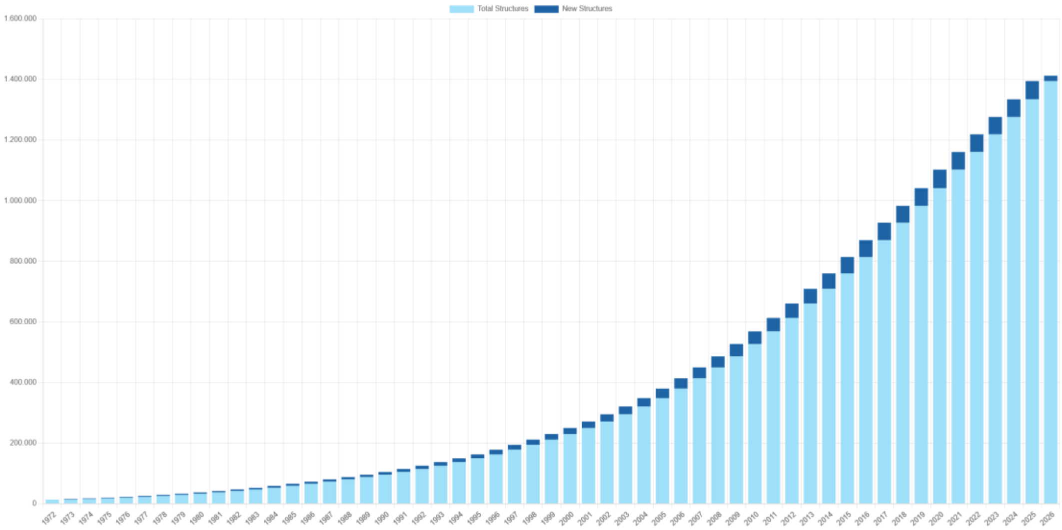
CSD growth

resulted in a rapid growth Crystal Engineering field

CSD 1972 to 2026

Δ ~ 55.000 - 65.000 entries per year





June 2026
1.454.146 entries
2025
~ 1.394.800 entries
2024
~ 1.334.700 entries
2023
~ 1.276.400 entries

2020 - 1 mln entries

What’s in the CSD?

Organic
43%

Metal-Organic
57%
At least one transition metal, lanthanide, actinide or any of Al, Ga, In, Tl, Ge, Sn, Pb, Sb, Bi, Po

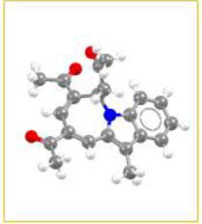
Not Polymeric
89%

Polymeric: 11%

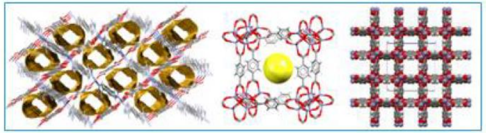
Single Component
56%

Multi Component
44%

Organic
- Drugs
- Agrochemicals
- Pigments
- Explosives
- Protein ligands



Metal-Organic
- Metal Organic Frameworks
- Models for new catalysts
- Porous frameworks for gas storage
- Fundamental chemical bonding



Additional data
- 10,860 polymorph families
- 169,218 melting points
- 840,667 crystal colours
- 700,002 crystal shapes
- 23,622 bioactivity details
- 9,740 natural source data
- > 250,000 oxidation states

Links/subsets
- Drugbank
- Druglike
- MOFs
- PDB ligands
- PubChem
- ChemSpider
- Pesticides

Where everything begun?... Supramolecular chemistry

Heinrich Heine Universität Düsseldorf

Molecule – focus of classical chemistry

- ultimate delimiter of all significant properties of a substance.
- only structures and functions involving covalent bond

BUT: some biological systems/phenomena do not involve making or breaking chemical bonds.

INSTEAD: bio systems are usually formed of loose aggregates held together by non-covalent interactions

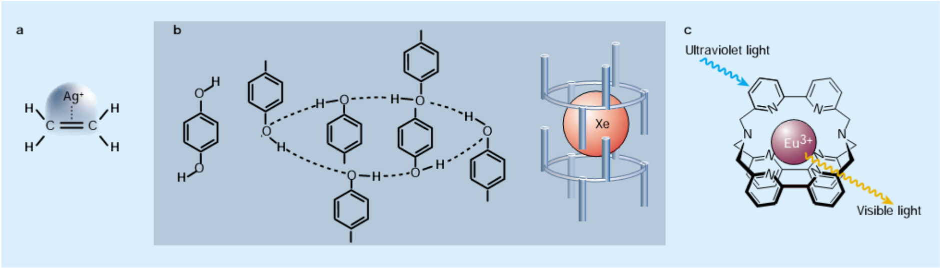
FURTHER: Some fundamental properties cannot be explained by molecular structure.

Supramolecular chemistry: main principles

Molecular recognition

- lock-and-key model
- enzyme-substrate interaction (E. Fischer 1894)
- molecules recognise one another through their dissimilarities rather than through their similarities

Chemistry beyond the molecule



G. Desiraju, *Nature*, (**2001**), *412*, 397.

Supramolecular function

- highly specific interactions lead to useful supramolecular functions

⇒ Interest - stable compounds that do not involve covalent bonds

Supramolecular chemistry is a chemistry of the intermolecular bond

J.-M. Lehn, 1969

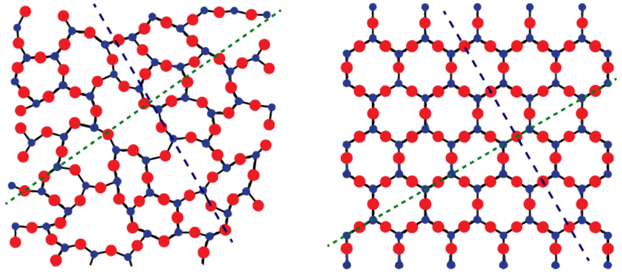
Figure: a. donor–acceptor b. clathrate using hydrogen bonds c. cryptand with metal ion

Crystal Engineering (CE) –

understanding of intermolecular interactions in the context of crystal packing

Aim: design of new solid materials with desired chemical and physical properties

Crystal is an <u>ordered</u> and <u>periodic</u> arrangement of molecules/atoms



design of solids with desired properties ↔ means design crystals where molecule are organised in certain ways.

What do we do?

→ make crystals by using interactions

How do we do it?

→ synthetic strategies

Why do we do it?

→ properties

<u>– operations in CE –</u>

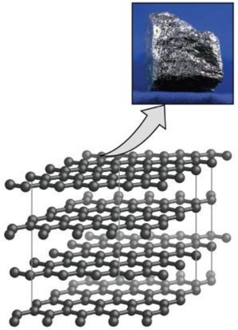
- Determination of crystal structures
- Understanding or analysis of crystal structures
- Use of understanding in trying to design a crystal structure
- Actual crystallization experiment
- Realization of pre-desired crystal property

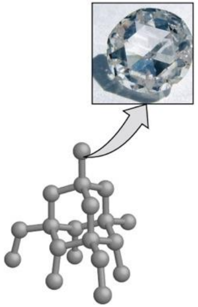
What is a molecular crystal?

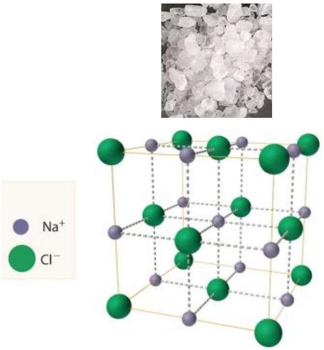
Heinrich Heine
Universität Düsseldorf

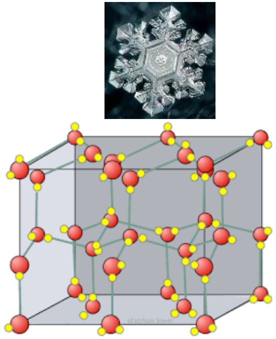
Molecular crystal is a **collection of molecules**, held internally by rather strong chemical bonds and associated with each other by weaker intermolecular interactions.

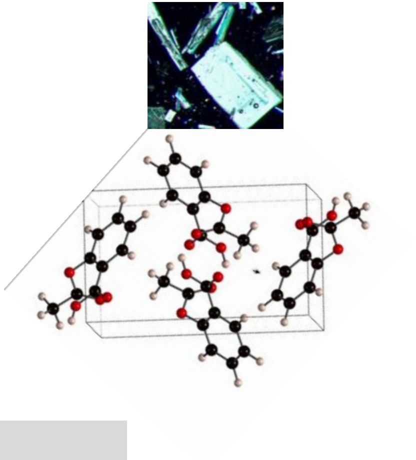
➢ molecular crystal is a crystalline form of any chemical substance that exists as molecule!











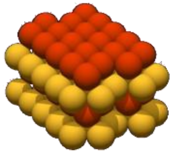
Fundamental questions of modern CE:
Given a molecular structure of a compound, what is its crystal structure?
How does crystalline structure correlate to the properties of the materials?

Understanding tools

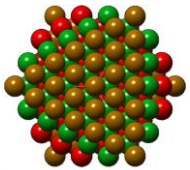
for aggregation phenomena of molecules

Close packing principle

molecules pack in a crystal so that any molecule is surrounded by a maximum number of close neighbors, generally 12.

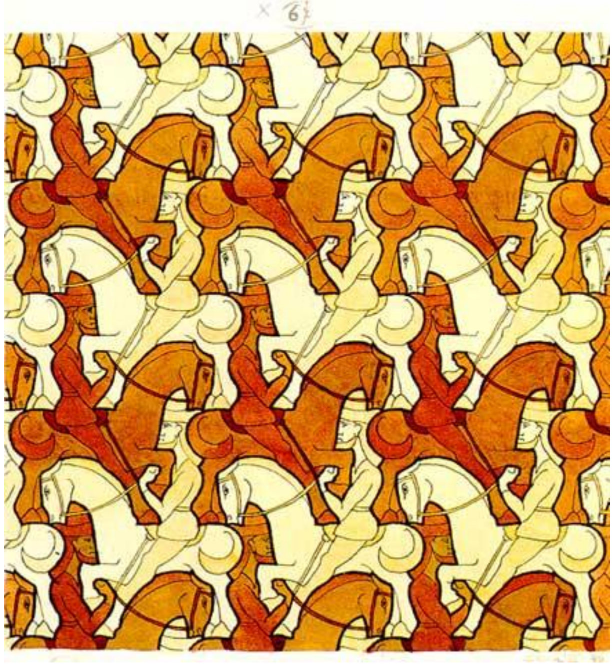


(hcp)



(ccp)

- minimization of the potential energy of the system
- isotropic attractive and repulsive forces between molecules
- highest possible number of stabilizing interactions



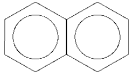
M.C. Escher's “Horsemen”

Understanding tools: close packing

Condensed aromatic hydrocarbons C_mH_n

naphthalene

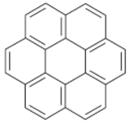
$C_{10}H_8$



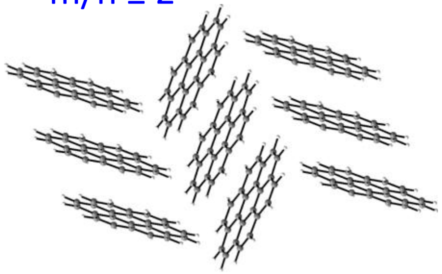
m/n < 1.5

coronene

$C_{24}H_{12}$



m/n ≥ 2



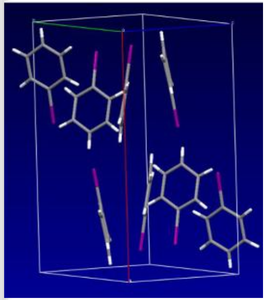
- hydrogen rich molecules
- angle ~40-50°
- herringbone geometry
- C-H··· π interactions

- carbon rich molecules (m/n ≥ 2)
- angle ~90°
- tendency toward π···π stacking

➢ C/H-stoichiometry is a critical parameter

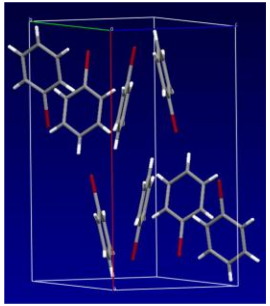
Further crystal packing similarity

I- and Br-derivatives – often „same“ packing



a=15.270 Å
b=11.071 Å
c= 7.703 Å
Pbcn

iodobenzene



a=14.264 Å
b=11.415 Å
c= 7.501 Å
Pbcn

bromobenzene

- „same“ packing
  → same arrangement on molecules
  → same cell parameters

➢ Isostructural compounds

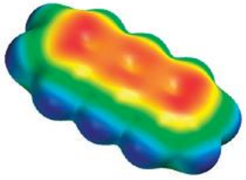
Pbcn – orthorhombic space group, all unit cell angles α=β=γ= 90 °; K. Merz, *Cryst.Growth Des.* **2006**, 6, 1615; S. G. Biswas, *Acta Crystallogr*. **1958**, 11, 882.

Understanding tools: beyond close packing

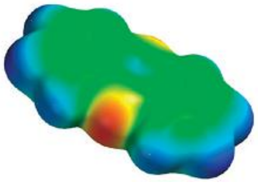
Heinrich Heine Universität Düsseldorf

Acenes vs N-Heteroacenes: The Effect of N-Substitution on the Structural Features of Crystals of Polycyclic Aromatic Hydrocarbons

Kenneth E. Maly



anthracene



phenazine

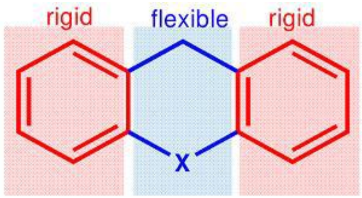
calculated electrostatic potential map

blue – positive, red - negative

- importance of C···H contacts is reduced
- C-H···N interactions in phenazine
- increased tendency to π···π stacking

➢ Heteroatom impact on crystal packing

Similarity as exception



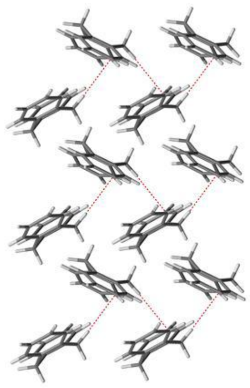
X = CH_2, NH

X = CH_2, dihydroanthracene (DHAN)

X = NH, dihydroacridine (DHAC)

DHAC

DHAN



- no significant impact of N-H group
- herringbone geometry
- similar C-H··· π and N-H··· π interactions
- isostructural compounds

➢ Protonation and absence of strong acceptors

K.E. Maly, *Cryst. Growth Des.*, **2011**, *11*, 5628-5633; A. Kupka, C. Schauerte, K. Merz, *Cryst. Growth Des.*, **2014**, *14(6)*, 2985.

Understanding tools

for aggregation phenomena of molecules

Intermolecular interactions

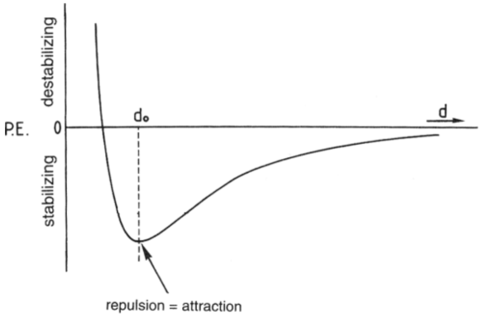
(between 2 atoms) ≤ ∑ of their van der Waals radii.

Strength > **weak and strong**
stabilizing & destabilizing **energies**

Distance > electrostatic and dispersive

Directionality > isotropic and anisotropic

Potential energy diagram for an intermolecular interaction



d_0 - equilibrium distance
→ lowest energy
→ repulsion = attraction

distances $d > d_0$ (and some shorter)
→ stabilizing $E < 0$
→ attractive forces

very short distances $d < d_0$
→ destabilizing $E > 0$
→ repulsive forces

stabilizing & destabilizing energies describe intermolecular interactions

Properties of intermolecular interactions

- Strength

  short/long & weak/strong

  distance & energy

Bond/ Interaction	Bond length (Å)
C-H	
C-C	
C=C	
C≡C	
C-O	
C=O	
O-H	
N-H	
O···H	

Bond/ Interaction	Bond energy (kJ/mol)
O-H	
C-H	
C-C	
C-N	
S-S	
Hydrogen bond	
Van der Waals (vdW)	~

Properties of intermolecular interactions

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- Strength

  short/long & weak/strong

  distance & energy

- majority of covalent bonds: 300 - 500 kJ/mol
- majority of hydrogen bonds: 4 - 60 kJ/mol

Many interactions important in context of crystal packing are 2-20 kJ/mol

→ energetically very weak!

- Intermolecular interaction: Distance < ∑ vdw radii

Properties of intermolecular interactions

Heinrich Heine
Universität Düsseldorf

- **Strength**
  weak and strong
  - majority of covalent bonds: 300 - 500 kJ/mol
  - majority of hydrogen bonds: 4 - 60 kJ/mol

- **Directionality**
  isotropic and anisotropic

isotropic interactions
non-directional
simple level of molecular recognition

→ responsible for supramolecular arrangements
→ close packing of the molecules
→ include vdW → vdW forces: attractive or repulsive
(between all atoms and molecules)

anisotropic interactions
directional
higher level of molecular recognition (lock-and-key principle)

→ determine intermolecular orientation and functions
→ involve partially charges atoms (N, O, Cl, P, S)
→ have electrostatic character – dipole-dipole interaction

→ master key – hydrogen bond – fits to different locks

Some important intermolecular interactions in CE

- hydrogen bonds (HB)
- halogen bonding (XB)
- $\pi\cdots\pi$ interactions (stacking)

Definition, importance & properties of H-bonds

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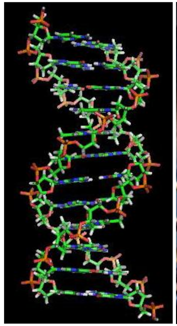
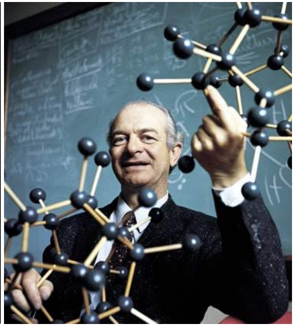
Master key of molecular recognition, **“the” toolkit** of CE

An attractive interaction **D–H···A–Z** between a hydrogen atom from molecular fragment D−H in which D is more electronegative than H, and an atom / group of atoms in the same or a different molecule. *Current IUPAC definition (2011)*





Linus Pauling

H is more electropositive than D (hydrogen bond donor) and A (HB acceptor)
→ **electrostatic character**

HB Properties

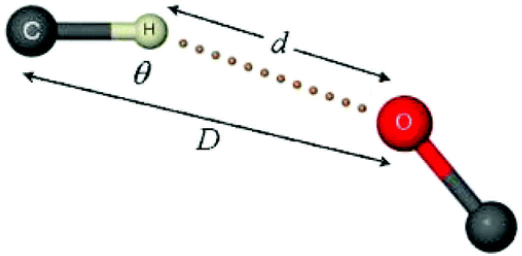
- Strength & length
- Directionality
- Cooperativity
- Hierachy

Gautam Desiraju

E. Arunan, G. R. Desiraju, R. A. Kleinet al, *Pure Appl. Chem.* **2011**, *83*, 1637.

Hydrogen bond, X–H···Y–Z

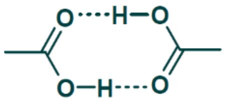
Strength	Example	X···Y (*D*, Å)	H···Y (*d*, Å)	X–H···Y (ϑ,°)	Bond energy (-kcal/mol)	Effect on crystal packing
very strong X–H ~ H···Y	$[F–H–F]^-$	2.2–2.5	1.2–1.5	175–180	15-40	strong
strong X–H < H···Y	O–H···O–H O–H···N–H N–H···O=C	2.6–3.0 2.6–3.0 2.8–3.0	1.6–2.2 1.7–2.3 1.8–2.3	145–180 150–180 150–180	4-15	distinctive
weak X–H << H···Y	C–H···O O–H···π	3.0–4.0	2.0–3.0	110–180 90–180	< 4	variable

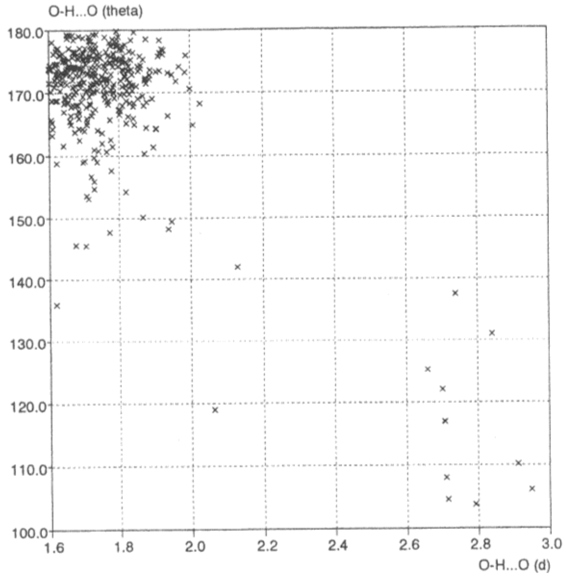


Cambridge Structural Database (CSD) analysis

Heinrich Heine
Universität Düsseldorf

O-H···O bonds in carboxylic acid dimers





strong HBs

- angle

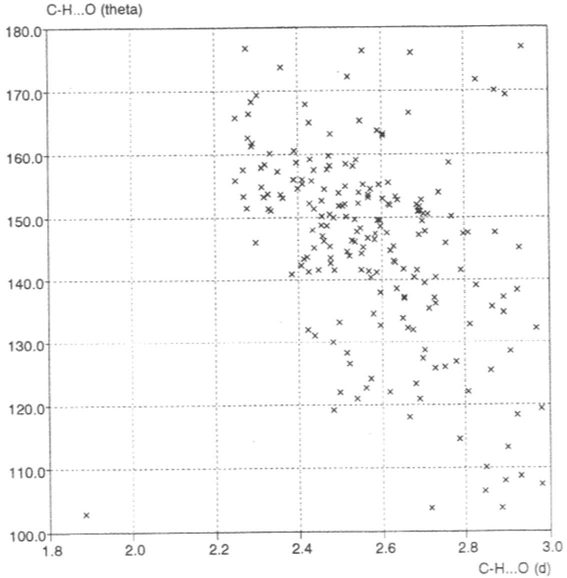
O-H...O: 160-180°

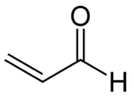
- distance

O...H : 1.6-1.9(2.0)Å

➢ well defined characteristics of strong H-bonds

C-H···O bonds in α,β-unsaturated carbonyl compounds





weak HBs

- angle

C-H...O: 100-180°

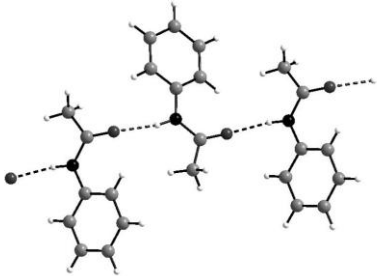
- Distance

O...H : 2.2-3.0 Å

➢ no information (or defined characteristics) about a general C-H...O H-bond possible

Angular properties of hydrogen bonds

Acetanilide



Linear N–H···O=C bonds connect secondary amide molecules in a one-dimensional array

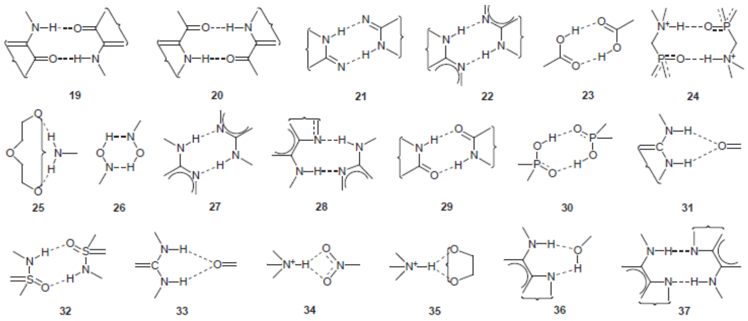
- chain-like arrangement
- two-centered interaction = “1-Point”HB
- linear angle D-H···A = 180 °

Diphenylurea

A bifurcated hydrogen bond, or three-centered interaction

- two equiv. donors, one acceptor
- non linear angle D-H···A < 180 °
- biological systems

Huge variety of hydrogen bonds motifs



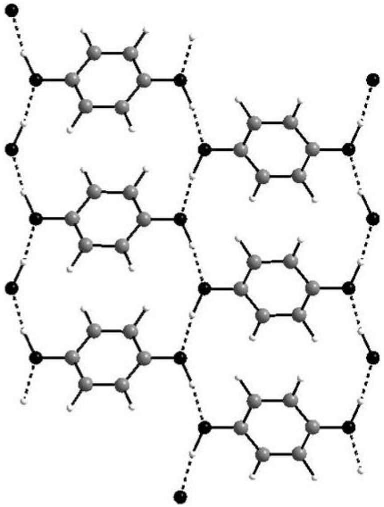
F. H. Allen, New J. Chem., (1999), 23, 25

Cooperativity of HBs

Heinrich Heine Universität Düsseldorf

e.g. alcohol **R-OH** (γ-hydroquinone):

acidity of H-atom in O–H···O–H dimer **>** in an isolated O–H group



HBs are more stable when involved in dimers, trimers, chains, infinite 2D- and 3D structures!

Why?

Polarizability & charge transfer effects:

total binding energy of all HB in an aggregate **> energy sum of the individual** HBs

Hierarchy of HBs

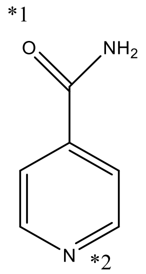
HB donors: OH, NH and NH_2

HB acceptors: **O**H, C=**O**, **N**H_2, -**N**= and –**O**-

In case of several HB donors and acceptors they pair up along their strength

⇒ the **best H-bond** is formed by the **best H-bond donor** and H-bond **acceptor**

⇒ the **second best HB**: by the **second best HB donor** and the second HB **acceptor**



Experiment:
mix isonicotinamide + two different aromatic acids (1:1:1)

*1:
*2:

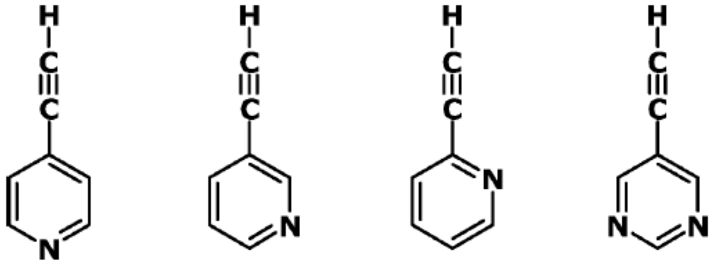
Hierarchy is a characteristic of kinetic (fast) crystallization

CE with weak HBs

Crystal Engineering with ≡C−H···N and =C−H···N Hydrogen Bonds

Venkat R. Thalladi,*,† Marta Dabros,† Annette Gehrke,‡ Hans-Christoph Weiss,‡ and Roland Boese*,‡

Crystal Growth & Design, Vol. 7, No. 4, 2007

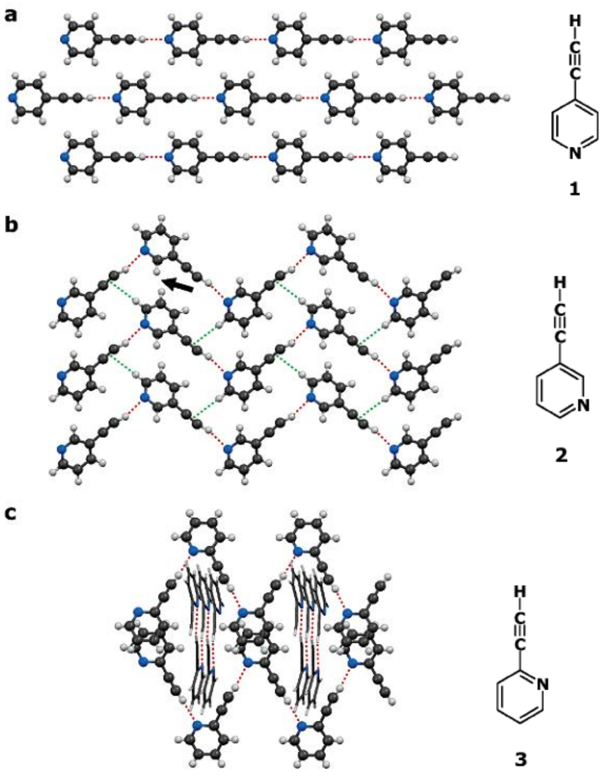


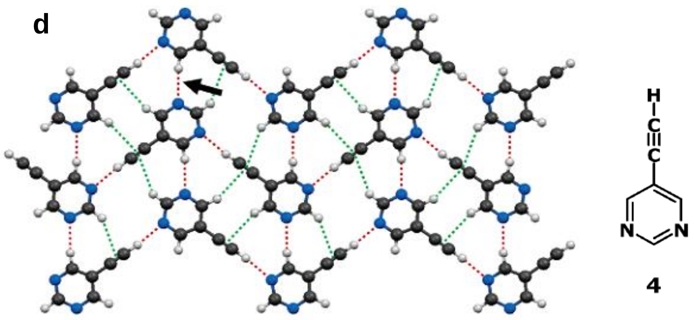
1 2 3 4

Ethynyl-pyridines and -pyrimidine

- assume that ≡C-H / =C-H groups form short & linear hydrogen bonds
- use weak HBs to generate networks with predictable topologies

Crystal Engineering with weak ≡C-H···N HBs





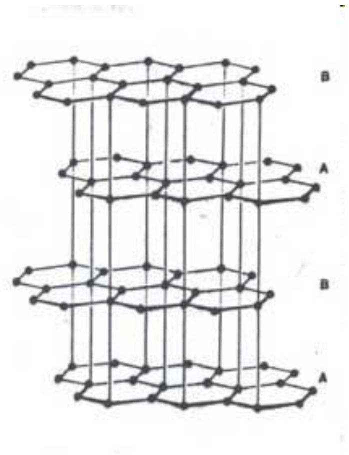
- straight (**1**) and zigzag (**2**,**3**,**4**) chain patterns formed by ≡C-H···N hydrogen bonds

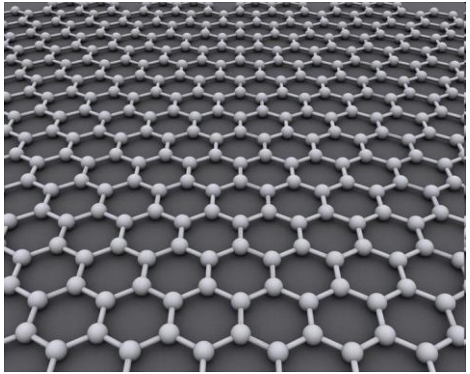
- additional cross-stitching of zigzag chains (**4**) by =C-H···N hydrogen bonds
→ 2D network

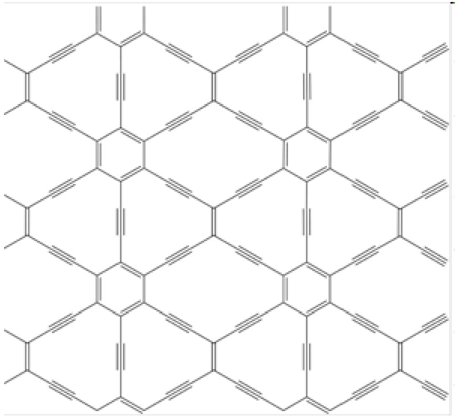
V. R. Thalladi et. al. *Crystal Growth & Design*, **2007**, *7*, 599.

Modulation of analoques of unknown network

Graphite | **Graphene** | **Graphyne**







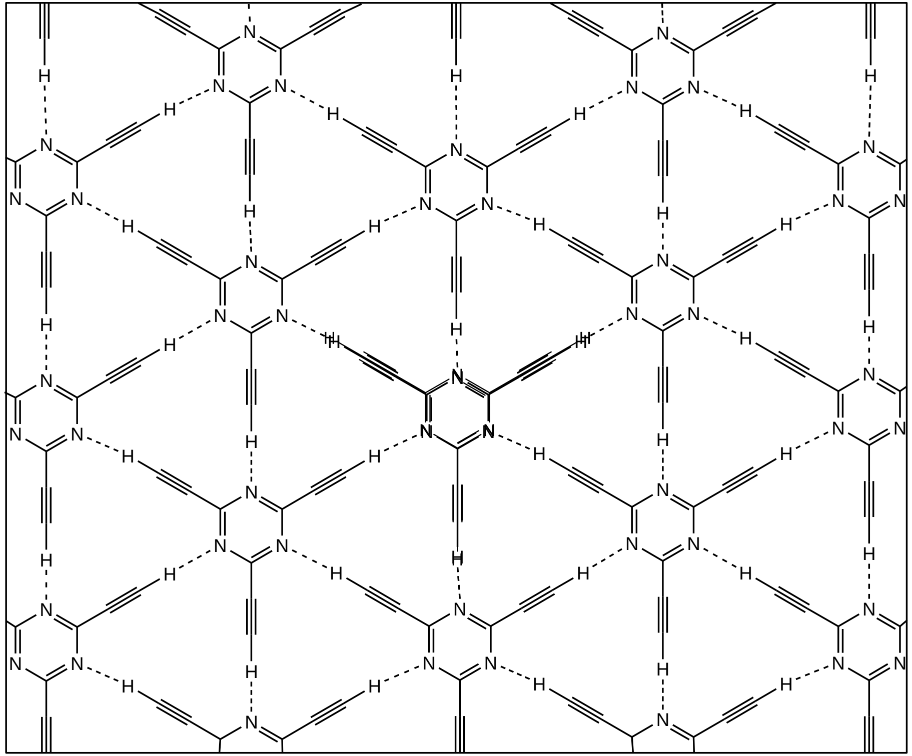
Graphene, a layer of graphite just one atom thick, isn't called a wonder material for nothing. The subject of the 2010 Nobel Prize in physics, it is famed for its superlative mechanical and electronic properties. Yet new computer simulations suggest that the electronic properties of a little-known sister material of graphene—graphyne—may in some ways be better.

The simulations show that graphyne's conduction electrons should travel extremely fast—as they do in graphene—but in only one direction. That property could help researchers design faster transistors and other electronic components that process one-way current, says one of the study's authors, theoretical chemist Andreas Görling of the University of Erlangen-Nuremberg in Germany. "If your material already conducts in one direction, you have a head start," he says.

J. Cartwright, *Sciene Now*, **01.03.2012**

Supramolecular graphyne:

robust 2D hydrogen-bonded network structure with suitable model: 2,4,6-triethynyl-1,3,5-triazine



- hexagonal array
- graphyne-like network
- 2D layers
- parallel stacking of layers
- weak C-H···N H-bonds
- <(C-H···N) = 180°
- all N and H atoms involved in the synthon formation

M. Ohkita, *Chem. Comm.* (**2002**) 3054

Building blocks for optoelectronic devices

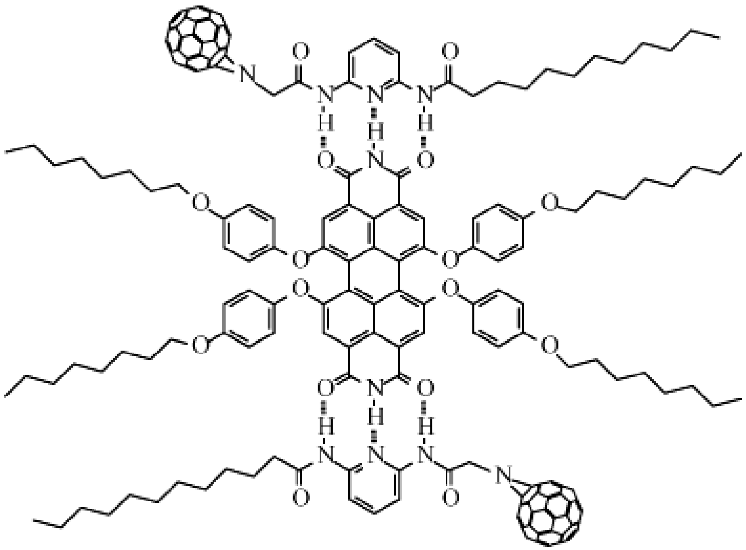
multiple hydrogen bonds for photovoltaic

Superstructure

Self-assembly of C_{60} fullerene derivative with PERY (perylene bisimide)

multiple HBs

DAD sequence



desired construction of functionalised fullerene-based architecture

Yang Liu, *J. Phys. Chem. B*, (**2004**), *108*, 6256.

Halogen Bonding (XB)…

Heinrich Heine
Universität Düsseldorf

SCIENTIFIC CONFERENCES

A big hello to halogen bonding

Halogen bonding connects a wide range of subjects — from materials science to structural biology, from computation to crystal engineering, and from synthesis to spectroscopy. The 1st International Symposium on Halogen Bonding explored the state of the art in this fast-growing field of research.

Mate Erdelyi

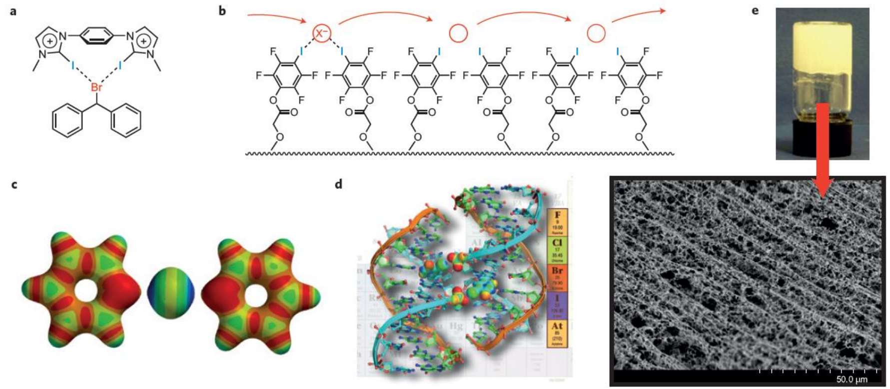


Figure 1 | Some highlights from the ISXB-1 meeting. **a**, A bidentate cationic halogen-bond donor designed for promoting halogen abstraction, for example in the solvolysis of benzhydril bromide. **b**, A synthetic ion channel that operates with halogen bonds. **c**, An iodonium ion is capable of forming two halogen bonds simultaneously, shown in the example of [bis(pyridine)iodine]$^+$. **d**, A DNA junction is an optimal model system to study biomolecular halogen bonds. **e**, An equimolar mixture of bis(pyridyl urea) and diiodotetrafluorobenzene in methanol-water forms a robust hydrogel upon rapid cooling. (Pui Shing Ho and Pierangelo Metrangolo are kindly acknowledged for providing figure parts **d** and **e**, respectively).

NATURE CHEMISTRY | VOL 6 | SEPTEMBER 2014 | www.nature.com/naturechemistry

IUPAC Definition

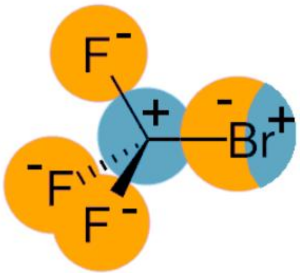
*“A **halogen bond** occurs when there is evidence of a net attractive interaction between an electrophilic region associated with a halogen atom in a molecular entity and a nucleophilic region in another, or the same, molecular entity”*

D-X···A **D-X···X-D**

(X = I, Br, Cl)
(A = O, N, S)

 G. R. Desiraju, P. S. Ho, P. Metrangolo, G.Resnati, K. Rissanen et al . *Pure Appl. Chem.* **2013**, *85*, 1711.

Nature of halogen bond



XB arise from polarization of C-X bond

⇒ X is polarized positively in the region furthest away from the C atom

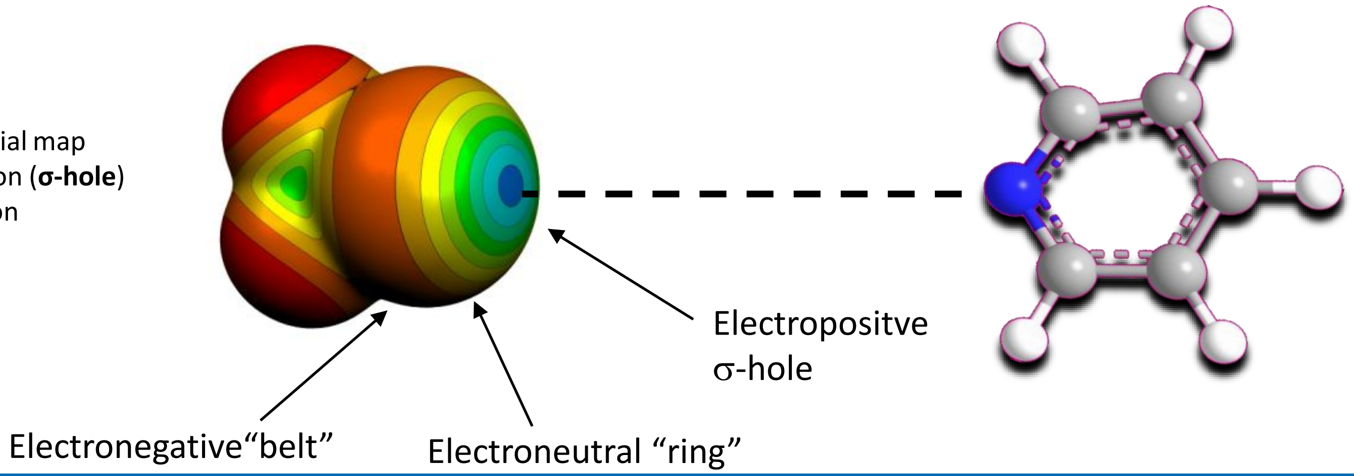
⇒ polarization gradient on C-X: $C^{\delta+}$- $X^{\delta-}$

⇒ Electrostatic contact with a negatively polarized species

⇒ Bond distance shorter than sum of VdW Radii

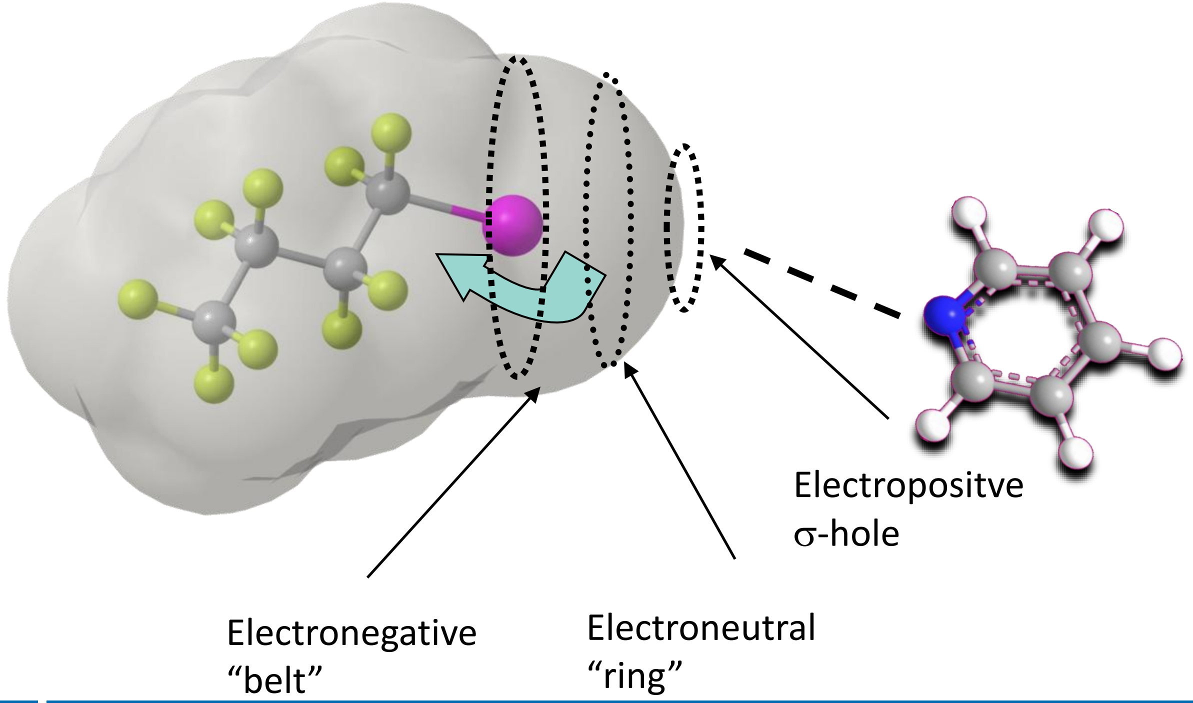
⇒ C-I···N angle approximately 180°

Electrostatic potential map
blue – positive region (**σ-hole**)
red – negative region



The σ-hole in halogen bonding

Perfluoroination: enlarges the σ-hole ⇒ interaction strength increases



- Polarization of C-X bond

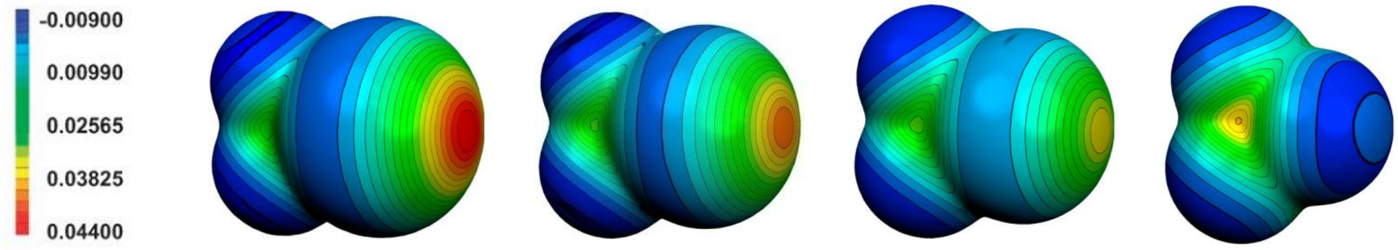
⇒ Electrostatic contact with a negatively polarized species

- Bond distance shorter than sum of VdW Radii
- **C-I···N** angle approximately 180°

The σ-hole dependence of halogen atom

Heinrich Heine Universität Düsseldorf

Perfluoroination: enlarges σ-hole ⇒ interaction strength increases



CF_3I CF_3Br CF_3Cl CF_4

Molecular electrostatic potential of CX_4 - halotrifluoromethane

Electrostatic potentials:
red – positive crown – **σ-hole**
blue – negative belt

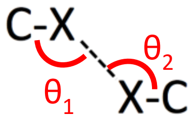
positive region (σ-hole)

- intermolecular interaction strength I > Br >> Cl > F
- interacts with a N, O, Halogen atom:
  C-I···N, C-I···O, C-Br···N, C-Br···N

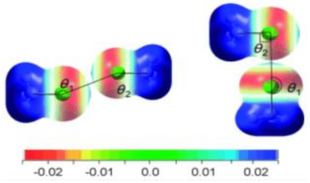
 T. Clark, et al. *J. Mol. Model.*, **2007**, *13*, 291. P. Metrangolo, G. Resnati et. al. , *Angew. Chem. Int. Ed.*, **2008**, *47*, 6114

Halogen···halogen interaction and it‘s geometry

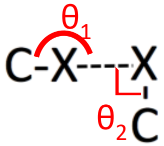
Calculated elektrostatic potentials
- blue – positive polarised,
- red – negatively polarised.



Type I



Type II



Geometry of two energy minimum of chloromethane

- symmetrical
- parallel, electrostatically identical regions
- van der Waals type
- shortest are actually repulsive
- $\theta_1 = \theta_2 \approx 150$
- Torsion angle (C–X···X–C), $\Phi = 180$

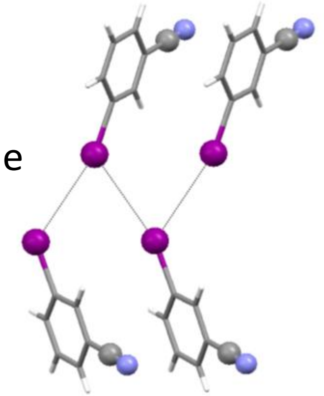
- genuine X-bond
- positive polarisation in the polar region of the halogen atom
- a negative polarisation in its equatorial region
- $\theta_1 = 180$ °und $\theta_2 = 90$ °

K. Merz, V. Vasylyeva, *CrystEngComm*, **2010**, *12*, 3989.

Properties of XB

Linear arrays in ***para***-iodobenzonitrile

meta-iodobenzonitrile
- zig-zag I···I chains



Merz, K., *Crystal Growth & Design*, **2006**, *6*, 1615.

→ Molecular recognition dependant on substitution pattern

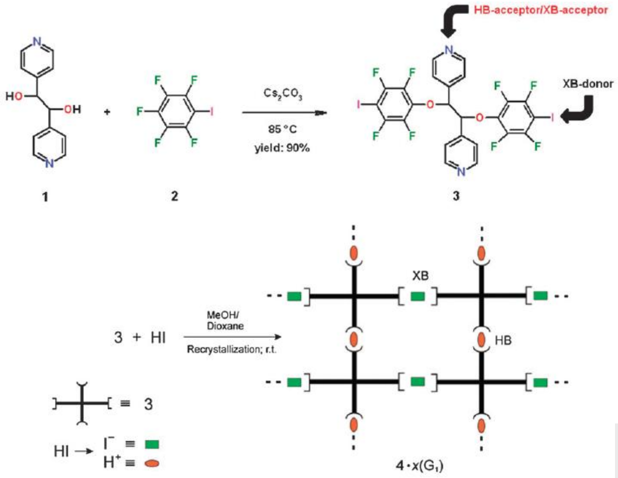
XB Properties

▪ directional	-- σ-hole
▪ tunable interaction strength	-- different depth of the σ-hole
▪ hydrophobic	-- negative and positive sides
▪ large donor atom size	-- tunable: I > Br > Cl

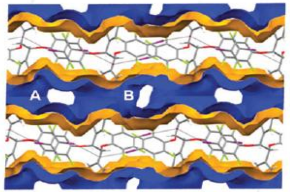
P. Metrangolo, G.Resnati *Halogen Bonding: Applications and Fundamentals*, Springer Verlag, **2008**.

Hydrogen and halogen bonding drive the orthogonal self-assembly of an organic framework possessing 2D channels†

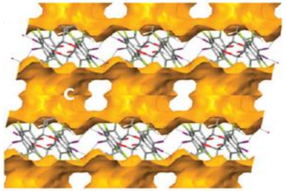
- open framework with 2D channels
- able to host various guest molecules
- single-crystal-to-single-crystal guest exchange



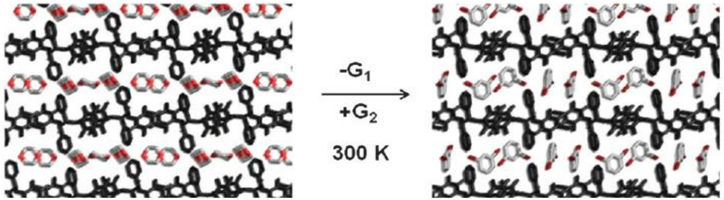
Voids observed after removal of guest molecules.



channels A and B



channel C intersects A and B channels



single-crystal-to-single-crystal guest exchange

XB a promising new tool for the design of open framework materials
→ pre-stage of non-covalently bonded metal-organic frameworks (MOF)

J. Marti-Rujas, L. Colombo, …, G. Resnati, *Chem. Commun.*, **2012**, *48*, 8207. ; contact surface – yellow;,solvent accessible surface – blue

XB in luminescent materials

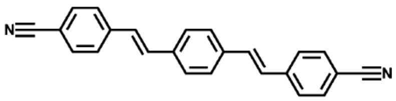
➢ self-assembling organic spin systems

➢ organic solid-state chromophores:

promising optoelectronic applications in the fields of light-emitting diodes, lasers, sensors, and two-photon fluorescent materials.

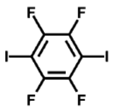
Challenge: obtain light-emission of an appropriate wavelength

fluorescent molecules **A**

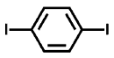


1,4-bis-*p*-cyanostyrylbenzene (A)

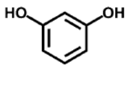
its co-formers **B-G**



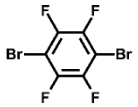
1,4-diiodotetrafluorobenzene (B)



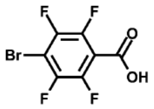
1,4-diiodobenzene (C)



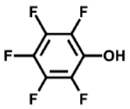
resorcinol (D)



1,4-dibromotetrafluorobenzene (E)



4-bromotetrafluorobenzene carboxylic acid (F)



2,3,4,5,6-pentafluorophenol (G)

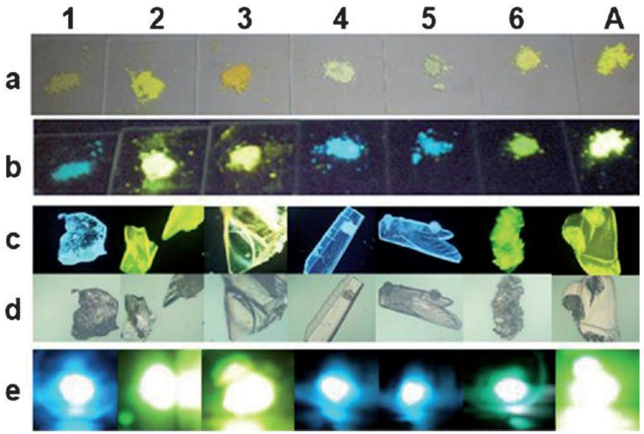


Figure 2. Photographs of solid-state cocrystal samples (from left to right: **1**–**6**) and **A**. a) and b) the powder samples under daylight and UV (365 nm); c) and d) the single crystal samples under UV (365 nm) and daylight observed by fluorescence microscope multiplied by 50-fold; e) two-photon luminescence under 800 nm laser.

Fine-tuning of UV/VIS absorption, luminescence emmision, color, lifetime, quantum yield and structural properties by cocrystal formation

XB in magnetic materials

➢ self-assembling organic spin systems

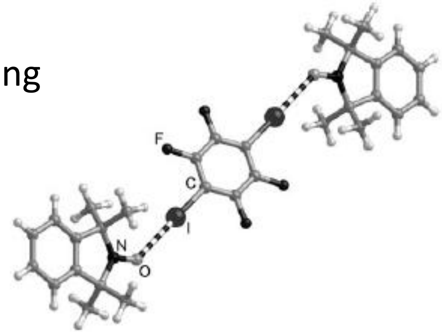
Heinrich Heine
Universität Düsseldorf

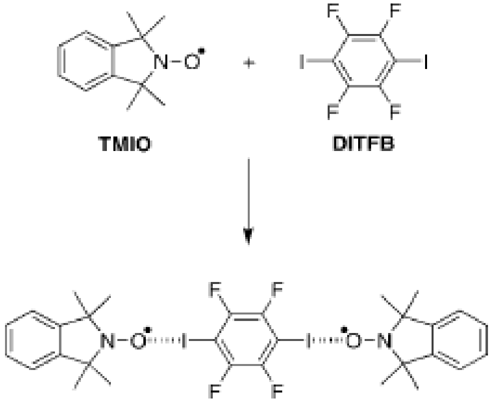
Iodoperfluorocarbons:

→ paramagnetic molecules as building blocks

Potential app.:

- molecular magnetism
- spintronics
- quantum computing





TMIO: tetramethylisoindolinyloxyl
DITFB: diiodotetrafluorobenzene

ionic resonance structure of the radical stabilized by N-O···I interaction

Maccaliff et. al, *Chem.Eur.J.*, **2009**, *15*, 4156

XB in fungicides

➢ case study of a preservative

3-Iodo-2-propynyl-*N*-butylcarbamate (**IPBC / Iodocarb**):

antimicrobial product used as preservative, fungicide and algaecide.

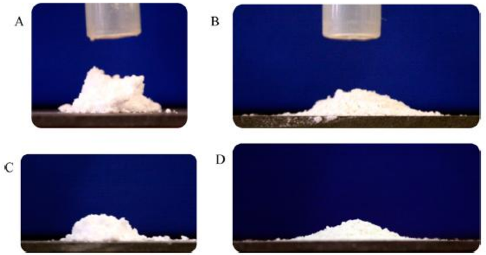


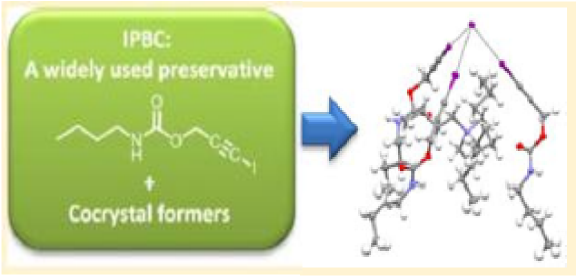
Figure 9. Pictures of cones of IPBC (A, C) and cocrystal 4 (B, D) powders, taken after flowing the powders through the funnel from 25 mm (A, B) and 50 mm (C, D) heights. The cylindrical shape of IPBC cones indicates clearly the high cohesion of the powders, while the flat cone shape of the cocrystal 4 indicates improved powder flow properties.

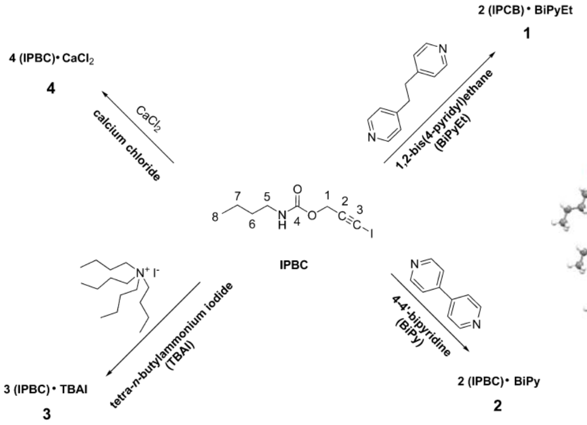
Problems:

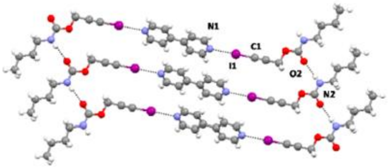
- Diffticult to obtain in pure form
- Hard to handle in industrial products and processes
- Sticky and clumpy

Idea:

- co-crystals







M. Baldrighi et al., *Mol. Pharmaceuticals*, **2013**, *10*, 1760

Fluorine substituent effect

Fluorine substituent effect

1. on physico-chemical properties ?
2. on interactions ?

Special properties of F:

- high electronegativity
- relatively small size
- very low polarisability

Perfluorocarbons

- Nearly ideal liquids.
- High compressibility.
- High vapor pressures.
- Poor solvents for organics.
  - non-polar molecules;
  - low polarizability;
  - small inter- and intra-molecular forces.

Hydrofluorocarbons

- Polar molecules.
- Appreciable inter- and intra-molecular interactions

Fluorine substituent effect

on physico-chemical properties of **fluorocarbons**:
only some tendencies

- Solubility (the least and the most polar solvents)
- Lipophilicity (increases in aromatic systems)
- Inductive effect (electron withdrawing)
- Acidity and basicity (increases HB acidity, decreases basicity)
- Steric effect (depends on compound and fluorination pattern)

Fluorine substituent effect

On interactions?

Fluorine: no σ-hole → no XB formation

What about HB?

F atom should be a stronger H-bond-acceptor than O or N

- correct for HF_2^-
- not correct for organic fluorinated compounds (C-F)

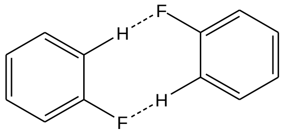
Table 1. Calculated hydrogen bond energies (kcal mol^{-1}) in some gas-phase dimers.[a]

Dimer	Energy
$[F{-}H{-}F]^-$	39
$[H_2O{-}H{-}OH_2]^+$	33
$[H_3N{-}H{-}NH_3]^+$	24
$[HO{-}H{-}OH]^-$	23
$NH_4^+\cdots OH_2$	19
$NH_4^+\cdots Bz$	17
$HOH\cdots Cl^-$	13.5
$O{=}C{-}OH\cdots O{=}C{-}OH$	7.4
$HOH\cdots OH_2$	4.7; 5.0
$N{\equiv}C{-}H\cdots OH_2$	3.8
$HOH\cdots Bz$	3.2
$F_3C{-}H\cdots OH_2$	3.1
$Me{-}OH\cdots Bz$	2.8
$F_2HC{-}H\cdots OH_2$	2.1; 2.5
$NH_3\cdots Bz$	2.2
$HC{\equiv}CH\cdots OH_2$	2.2
$CH_4\cdots Bz$	1.4
$FH_2C{-}H\cdots OH_2$	1.3
$HC{\equiv}CH\cdots C{\equiv}CH^-$	1.2
$HSH\cdots SH_2$	1.1
$H_2C{=}CH_2\cdots OH_2$	1.0
$CH_4\cdots OH_2$	0.3; 0.5; 0.6; 0.8
$C{=}CH_2\cdots C{=}C$	0.5
$CH_4\cdots F{-}CH_3$	0.2

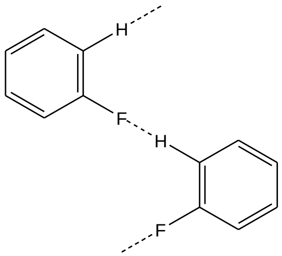
Fluorine substituent effect

On Interactions?

- C-H···F interactions similar to C-H···O und C-H···N, but very weak
- C-H···F well established but pretty unpredictable, system dependent



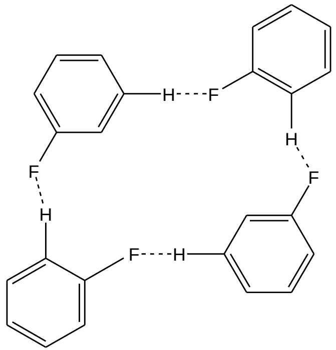
I



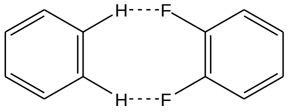
II

and still…

Classification of packing motifs with C-F···H interactions



III



IV

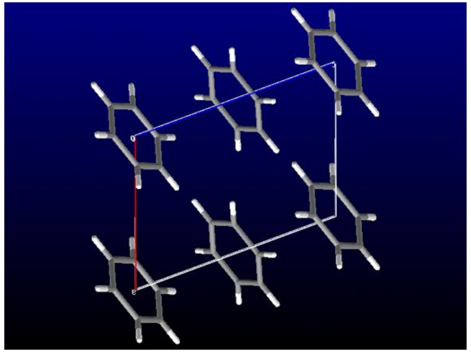
C-H···F comparable to C-H···N hydrogen bonds

➢ stabilizing specific crystal structures

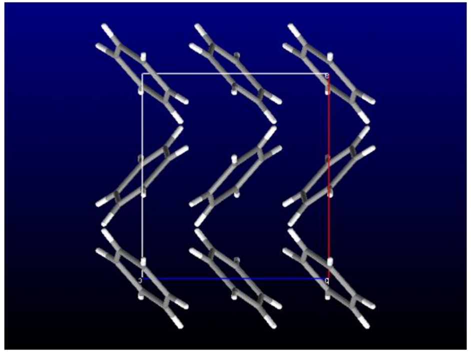
V. Vasylyeva, K. Merz *J. Fluorine Chem.*, **2010**, *131*, 446

Crystal packing of benzene

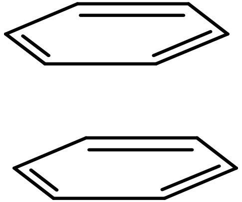
- isostructural to fluorobenezene C_6F_6



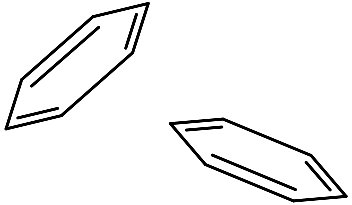
high pressure modification



modification at 1 atm



face-to-face π···π stacking

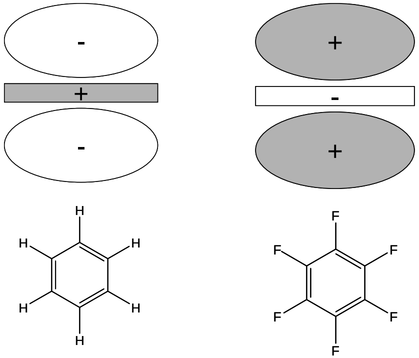


edge-to-face C-H···π interaction

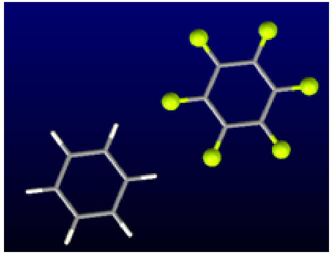
Crystal Engineering with C_6H_6 and C_6F_6

Quadrupol moments:

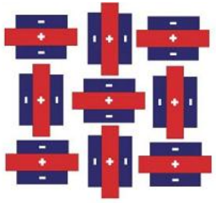
C_6H_6 C_6F_6



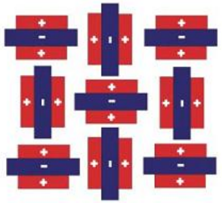
benzene hexafluorobenzene

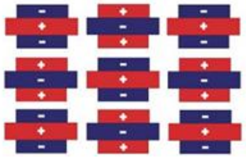


C_6H_6

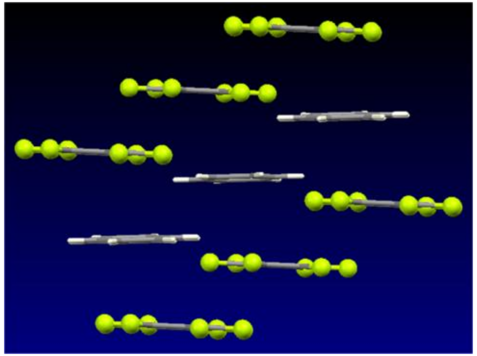


C_6F_6





C_6H_6 x C_6F_6 co-crystal



π···π stacking

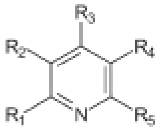
→ aryl-perfluoroaryl interaction

Aryl-perfluoroaryl interactions in CE co-crystal formation

➢ XB + aryl-perfluorarylstacking

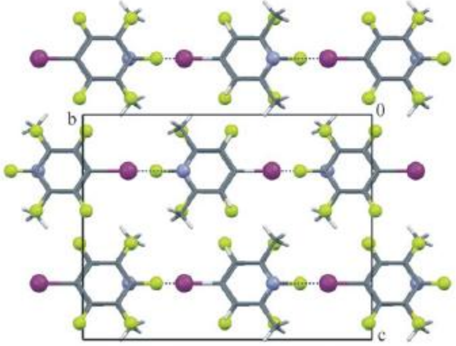
Heinrich Heine Universität Düsseldorf

C_6F_5I

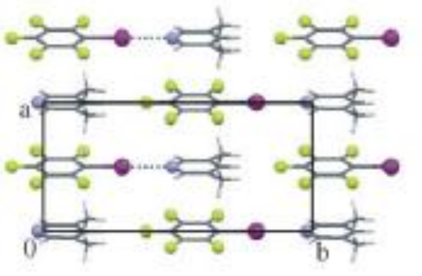


1

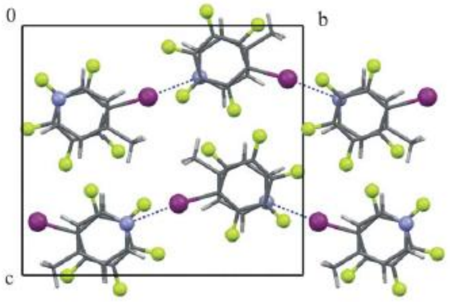
2 $R_1 = R_3 = R_5 =$ Me; $R_2 = R_4 =$ H
3 $R_1 = R_5 =$ Me; $R_2 = R_3 = R_4 =$ H
4 $R_2 = R_4 =$ Me; $R_1 = R_3 = R_5 =$ H
5 $R_3 =$ Me; $R_1 = R_2 = R_4 = R_5 =$ H



(a)



(c)



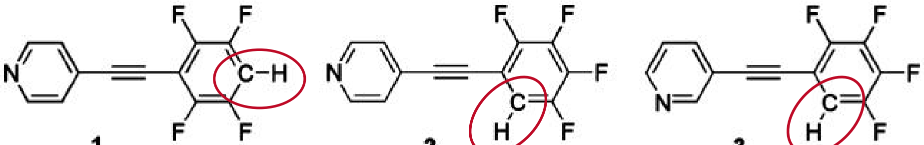
(b)

Polonski et. al, CrystEngComm, 2007, 9, 203–206

Aryl-perfluoroaryl interactions in CE

Heinrich Heine
Universität Düsseldorf

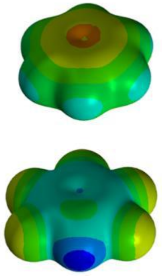
- HB + aryl-perfluorarylstacking
- weak C-H···N HBs in F-substituted phenylethynylpyridines



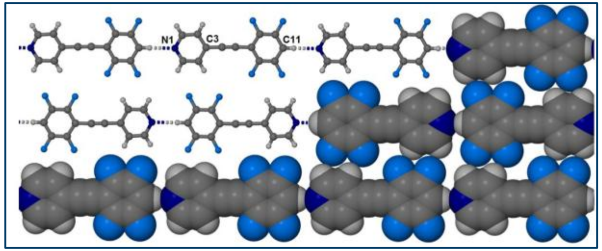
1 2 3

→ Increased acidity of phenyl proton by fluorination of the benzene ring

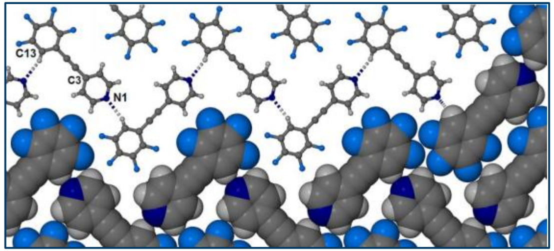
Electrostatic potential map of C_6H_6 and C_6HF_5



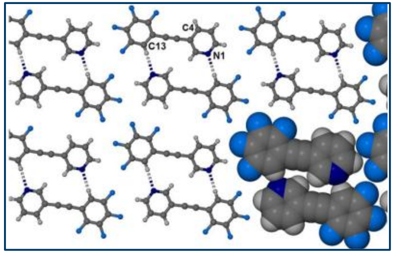
blue – positive charge



1: linear chains



2: zig-zag chains



3: dimers

E. Bosch, N. P. Bowling, J. Darko, *Crystal Growth & Design*, **2015**, *15*, 1634.

Fluorine applications in coatings and theranoctics

Linear and branched perfluoropolyether coatings on diamond-like carbon films: covalent linkage and physical deposition

WALTER NAVARRINI[1], MAURIZIO SANSOTERA[1*], FRANCESCO VENTURINI[1], CLAUDIA L. BIANCHI[2], ANTONIO GUARDA[3], GIUSEPPE RESNATI[1]

Application of liquid lubricant films on diamond-like carbon (DLC) coatings.
Hard surfaces are typically lubricated with perfluoropolyethers (PFPE).

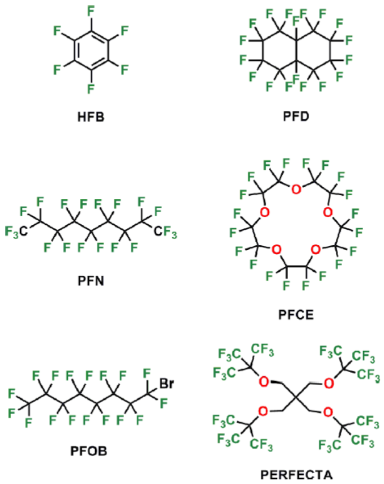
Water droplets on several DLC **coatings**

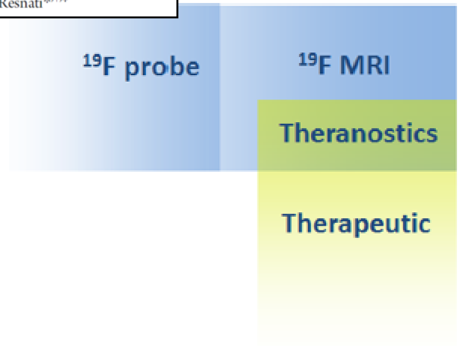
blank DLC thin film | linear PFPE coating | branched PFPE coating

CHEMICAL REVIEWS
Review
pubs.acs.org/CR

^{19}F Magnetic Resonance Imaging (MRI): From Design of Materials to Clinical Applications

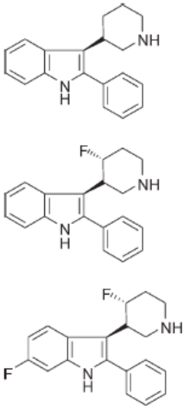
Ilaria Tirotta,[†,‡] Valentina Dichiarante,[†,‡] Claudia Pigliacelli,[†,‡] Gabriella Cavallo,[†,‡] Giancarlo Terraneo,[†,‡] Francesca Baldelli Bombelli,[*,†,‡,§] Pierangelo Metrangolo,[*,†,‡,∥] and Giuseppe Resnati[*,†,‡]



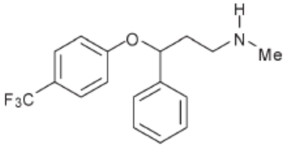


Polyfluorocarbons as **MRI tracers**: molecular tracers, polymers, branched derivatives

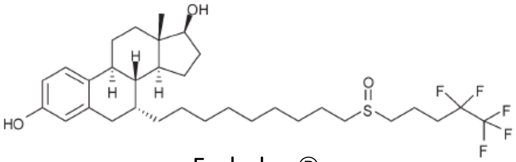
Fluorine in medicinal chemistry



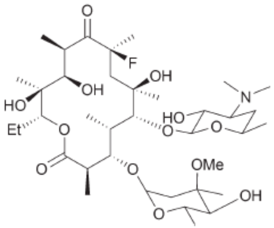
3-piperidinylindole
antipsychotic drugs



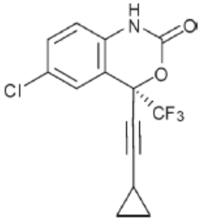
Prozac ®
antidepressant



Faslodex ®
oestrogen receptor antagonist



Flurithromicin
antibiotic



Efavirenz
transcriptase inhibitor HIV treatment

In a series of 3-piperidinylindole: fluorination decreased the basicity of the amine, thereby improving bioavailability